

Enhancing Security in GCC Fintech: Framework, Policies and Industry Standards

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Abstract

This white paper explores the critical aspects and proposes a framework for enhancing security in the fintech sector within the Gulf Cooperation Council (GCC) region. It leverages existing best practices and insights from the literature and organisations. Firstly, the study investigates and assesses the current regulatory frameworks, identifies weaknesses, and underscores the challenges posed by inconsistent regulations, high compliance costs and skills gaps across GCC countries. Then, it highlights the significance of robust cybersecurity measures, the integration of emerging technologies such as blockchain and artificial intelligence, the necessity for continuous collaboration between fintech companies, regulators, and cybersecurity experts, and suggests industry standards and best practices which are useful for fostering a secure and efficient fintech environment. By attempting to propose an initial framework that combines regulatory needs, identified weaknesses, and industry standards, this research aims to provide actionable recommendations for policymakers, industry stakeholders, and academia. The findings emphasise the importance of a harmonised regulatory approach, improved technological infrastructure, and specialised policies as education programs to support the sustainable growth of Fintech in the GCC. The paper concludes with a discussion of the study's limitations and suggests directions for future research.

CCS Concepts

• **Information systems** → Information systems applications; Enterprise information systems.

Keywords

Fintech Security, Performance Standards, GCC Region, Blockchain Applications, Data Management

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1 Introduction

The Fintech industry within the Gulf Cooperation Council (GCC) region has gained a lot of traction and is experiencing unprecedented growth, driven by technological advancements and a favourable regulatory environment [1] (Tsanis, 2021). As Fintech startups proliferate and traditional financial institutions increasingly adopt digital solutions, ensuring robust security and optimal performance has become a critical priority. This paper aims to comprehensively analyse the current security and performance standards in the GCC Fintech sector, identifying key challenges and opportunities for improvement.

The rationale for this research stems from the need to address the heightened risk of cyber threats and performance bottlenecks accompanying rapid innovation in the Fintech landscape. The emergence of financial technology (FinTech) in GCC has transformed traditional banking and regulatory systems from risky behaviour or other potential disruptions (Khan et al., 2022). Nevertheless, while FinTech is significantly growing in the Gulf Cooperation Council (GCC) region, it is not as developed as in other developed countries like China and the United States and critical factors and barriers are preventing that [2] (Khan et al., 2022).

Cybersecurity can be considered a main challenge, as well as properly managing risks for integrating FinTech services, since cyber-attacks are increasingly threatening FinTech [3] (AlBenjasim et al., 2023). At the same time, Fintech products and services improve the performance of financial institutions [4] (Chen et al., 2023). Technology, specifically Financial Technology, can improve performance in various ways, such as increasing efficiency, reducing costs, including compliance costs, and offering innovative services and products that generate new revenue streams. Performance, mainly expressed by profitability measures, is a key business objective. Therefore, it can be argued that cybersecurity and performance are two key challenges for FinTech development. This paper is an initial effort to create a framework for facilitating and enhancing cybersecurity and performance with specific applications to the GCC context.

This study advances knowledge for academic and practice communities by offering new and more recent insights. It aids in curriculum development, ensuring that educational institutions can prepare students for careers in the evolving FinTech sector, and fosters opportunities for collaborative research projects across various disciplines. For industry professionals, this research identifies best practices and standards to enhance the security and performance of Fintech solutions. Companies can implement more effective risk management strategies by understanding the prevalent cybersecurity threats and performance challenges. The insights gained from

this study can drive innovation, enabling companies to develop more secure and effective Fintech products, thereby maintaining a competitive edge in the market.

Governmental bodies can also benefit from this research through its evidence-based recommendations that inform the development of regulatory frameworks to secure the FinTech sector. Understanding the current landscape helps ensure that existing regulations are effective and that new policies address emerging threats and challenges. Strengthening cybersecurity and performance standards in Fintech contributes to increased public confidence in digital financial services, supporting broader economic goals.

Other stakeholders, including investors, can use the findings of this research to guide their investment decisions by highlighting key areas of risk and opportunity within the Fintech sector. Enhanced security and performance standards directly benefit consumers by protecting their financial data and ensuring the reliability and integrity of financial services.

2 Literature

2.1 Literature Overview and Methods

The rapid expansion of the fintech sector in the Gulf Cooperation Council (GCC) region has brought significant innovation opportunities and substantial cybersecurity challenges. Understanding various challenges, such as regulatory frameworks, adoption barriers, and cybersecurity measures, is crucial for stakeholders across the fintech ecosystem. This literature overview synthesises findings from various studies, offering a critical and interconnected analysis of Fintech and cybersecurity in the GCC, which will be used later to generate the proposed framework.

[2] Khan et al. (2022) provide a systematic literature review and extensive mapping identifying barriers for FinTech in GCC countries, and they find that some of the key problems are uncertainty primarily related to international standards and technological changes to legal systems; elevated risk, mainly cyber-attacks; as well as costs concerning the mitigation of risks and compliance (i.e. KYC/CDD checks). Based on this mapping study and related literature, we focus on regulatory challenges and uncertainty cybersecurity concerns and expand on financial inclusion and sustainability, which are United Nations goals. The institutional setting and, therefore, regulations differ across GCC countries. As one of the key challenges remains cybersecurity, the paper presents an analysis of the GCC context and the cybersecurity authorities in the GCC region and an overview of Fintech regulations. Then, the international perspective is presented, mainly focusing on the World Bank and international standards, such as ISACA industry best practices and policies. Based on this literature overview, we identify weaknesses in the GCC countries and international standards and propose a framework to improve cybersecurity in the following section.

2.1.1 Regulatory Frameworks and Cybersecurity. Several studies highlight the critical role of regulatory frameworks in managing the cybersecurity risks associated with fintech innovations. [5] Khan, Khan, and Ghafoor (2023) emphasise that while fintech adoption can enhance efficiency and customer experience, it also introduces significant cybersecurity threats. Regulatory environments, such

as those implemented by the Central Bank of Bahrain (CBB), play a pivotal role in mitigating these risks through robust cybersecurity measures, including risk assessments and data protection policies [6] (Alghamdi & Elmedany, 2023).

[7] Allen (2021) expands on this by examining the broader MENA region, noting that regulatory frameworks must evolve to address the unique risks of digital currencies and blockchain technologies. As suggested by [5] Khan et al. (2023), the integration of regulatory sandboxes allows for innovation while managing potential risks, highlighting a common theme across the studies on the importance of balancing innovation with regulatory oversight.

2.1.2 Cybersecurity Concerns and Adoption Barriers. The adoption of Fintech in the GCC is influenced by various barriers, including regulatory uncertainties and cybersecurity concerns. [8] Al Rahma and Al-Alawi (2023) identify key success factors for fintech startups, such as strong management teams and robust cybersecurity measures. Their findings align with those of [9-10] Abdeldayem and Al Dulaimi (2020), who emphasise the necessity for comprehensive cybersecurity education within fintech curricula to better prepare future professionals for the evolving digital landscape.

The studies also highlight the cultural and operational challenges in adopting fintech solutions. Allen (2021) and [8] Al Rahma and Al-Alawi (2023) both point to the need for tailored evaluation criteria and systematic approaches to overcome these barriers. The importance of cybersecurity as a competitive advantage is a recurring theme, underscoring the necessity for fintech firms to adopt best practices and comply with international cybersecurity standards like ISO/IEC 27001 and PCI-DSS [6](Alghamdi & Elmedany, 2023).

2.1.3 Impact on Financial Inclusion and Sustainability. [11] Alquliti (2022) provides a unique perspective by exploring the impact of Fintech, green finance, and financial inclusion on energy efficiency in the GCC. While financial inclusion and green financing significantly influence energy efficiency, the impact of Fintech is less pronounced. This finding suggests that while Fintech has the potential to drive financial inclusion, its implementation must be carefully managed to ensure broader sustainability goals.

The integration of Fintech with Islamic finance principles presents additional challenges and opportunities. Abdeldayem, Al Dulaimi, and Nekhili (2020) explore the compatibility of cryptocurrencies with Islamic finance, revealing a lack of consensus among scholars. This underscores financial and Sharia authorities' need for dual regulation to ensure ethical and legal standards compliance.

2.1.4 Practical Implementations and Future Directions. Practical implementations of fintech solutions highlight the diverse approaches within the GCC. The studies collectively suggest that collaboration between fintech companies, regulators, and cybersecurity experts is crucial for enhancing the overall security posture of the sector. The potential of emerging technologies, such as blockchain and artificial intelligence, to improve cybersecurity is a promising avenue for future research and development ([7] Allen, 2021; [6] Alghamdi & Elmedany, 2023).

As proposed by Abdeldayem and Al Dulaimi (2020), integrating cybersecurity into fintech education is essential for developing

Country	Authority Name
Bahrain	National Cyber Security Centre (NCSC)
Kuwait	Central Agency for Information Technology (CAIT)
Kuwait	Communication and Information Technology Regulatory Authority (CITRA)
Oman	Information Technology Authority (ITA) - National Centre for Information Security (NCIS)
Qatar	Qatar National Cyber Security Agency (NCSA)
Qatar	Ministry of Transport and Communications - Cyber Security Division
Saudi Arabia	National Cybersecurity Authority (NCA)
Saudi Arabia	Saudi Communications and Information Technology Commission (CITC)
UAE	Telecommunications and Digital Government Regulatory Authority (TDRA)
UAE	National Electronic Security Authority (NESA)

a skilled workforce capable of addressing the cybersecurity challenges in the fintech sector. This approach aligns with the need to continuously monitor and update cybersecurity practices to protect against evolving threats [5] (Khan et al., 2023).

2.2 The GCC Context

2.2.1 *Cybersecurity Authorities in the GCC Region.* The Gulf Cooperation Council (GCC) countries have established dedicated cybersecurity authorities to oversee and enhance the security of their digital infrastructures. These authorities play a crucial role in formulating policies, implementing regulations, and responding to cyber threats. They also collaborate with international organisations and industry stakeholders to ensure a robust cybersecurity framework. Below is an overview of the key cybersecurity authorities in each GCC country:

Bahrain: In Bahrain, the National Cyber Security Centre (NCSC) safeguards the nation’s critical information infrastructure. The NCSC develops cybersecurity policies, conducts awareness campaigns, and provides guidance on best practices to protect against cyber threats.

Kuwait: The Central Agency for Information Technology (CAIT) and the Communication and Information Technology Regulatory Authority (CITRA) spearheaded Kuwait’s cybersecurity efforts. These bodies focus on enhancing cybersecurity resilience, promoting secure information exchange, and enforcing cybersecurity regulations nationwide.

Oman: The Information Technology Authority (ITA), through its National Centre for Information Security (NCIS), leads Oman’s cybersecurity initiatives. The ITA/NCIS develops and enforces cybersecurity standards, conducts training programs, and collaborates with regional and international partners to strengthen cybersecurity measures.

Qatar: Qatar has established the National Cyber Security Agency (NCSA) and the Ministry of Transport and Communications’ Cyber Security Division to address its cybersecurity needs. These entities are tasked with developing national cybersecurity strategies, monitoring cyber threats, and ensuring critical infrastructure protection.

Saudi Arabia: The National Cybersecurity Authority (NCA) and the Saudi Communications and Information Technology Commission (CITC) are the main cybersecurity bodies in Saudi Arabia. They are responsible for creating a secure and resilient cyberspace,

implementing national cybersecurity policies, and coordinating responses to cyber incidents.

United Arab Emirates: The UAE’s Telecommunications and Digital Government Regulatory Authority (TDRA) and the National Electronic Security Authority (NESA) oversee the nation’s cybersecurity landscape. These organisations develop cybersecurity regulations, promote awareness, and work towards enhancing the UAE’s cyber resilience.

2.2.2 *Fintech Regulations in the GCC.* The regulatory landscape for Fintech in the Gulf Cooperation Council (GCC) countries is evolving rapidly, reflecting the region’s ambition to become a global hub for financial technology. Each GCC country has taken significant steps to create a conducive regulatory environment that balances innovation with security and consumer protection. This section provides an overview of the key regulatory frameworks and initiatives in the GCC, highlighting the approaches taken by each country to foster the growth of Fintech while ensuring robust oversight.

Bahrain: Bahrain has established itself as a pioneer in Fintech regulation within the GCC. The Central Bank of Bahrain (CBB) has implemented a comprehensive Fintech regulatory sandbox, allowing startups to test their innovations in a controlled environment. This initiative supports the development of new technologies while ensuring regulatory compliance. Additionally, the CBB has issued specific regulations for crowdfunding, digital financial advice, and cryptocurrency exchanges, positioning Bahrain as a forward-thinking jurisdiction in the Fintech space.

Kuwait: Kuwait’s approach to Fintech regulation is spearheaded by the Central Bank of Kuwait (CBK) and the Communication and Information Technology Regulatory Authority (CITRA). These bodies have introduced regulatory sandboxes to facilitate innovation in financial services. The CBK has also focused on digital payments and cybersecurity, issuing guidelines that promote secure and efficient digital transactions. Ongoing efforts aim to modernise the financial sector by adopting advanced technologies and regulatory practices.

Oman: In Oman, the Central Bank of Oman (CBO) and the Capital Market Authority (CMA) oversee Fintech regulations. The CBO has introduced a regulatory framework that supports electronic payments, mobile banking, and blockchain technology. Oman’s regulatory sandbox allows Fintech startups to experiment with new products under the supervision of regulators. The CMA has also

Country	Key Regulatory Bodies	Major Initiatives
Bahrain	Central Bank of Bahrain (CBB)	Fintech regulatory sandbox, cryptocurrency regulations, digital financial advice
Kuwait	Central Bank of Kuwait (CBK), CITRA	Regulatory sandbox, digital payments guidelines, cybersecurity regulations
Oman	Central Bank of Oman (CBO), Capital Market Authority (CMA)	Regulatory sandbox, electronic payments framework, blockchain technology support
Qatar	Qatar Central Bank (QCB), Qatar Financial Centre (QFC)	Regulatory sandbox, electronic payments regulations, digital wallets guidelines
Saudi Arabia	Saudi Arabian Monetary Authority (SAMA), Capital Market Authority (CMA)	Regulatory sandbox, digital financial services guidelines, Vision 2030 initiatives
UAE	Central Bank of the UAE (CBUAE), Dubai Financial Services Authority (DFSA), Abu Dhabi Global Market (ADGM)	Fintech regulatory frameworks, digital payments regulations, electronic KYC processes

been active in regulating digital assets and crowdfunding platforms, promoting a secure and innovative financial ecosystem.

Qatar: Qatar has made significant strides in developing a Fintech-friendly regulatory environment. The Qatar Central Bank (QCB) has launched a regulatory sandbox to encourage Fintech innovation. Additionally, the Qatar Financial Centre (QFC) provides a legal and regulatory framework that supports a wide range of financial services, including Fintech. The QCB has also issued regulations to govern electronic payments, digital wallets, and cybersecurity, ensuring a robust infrastructure for digital finance.

Saudi Arabia: Saudi Arabia's Vision 2030 initiative has been driving the country's Fintech regulatory advancements. The Saudi Arabian Monetary Authority (SAMA) and the Capital Market Authority (CMA) have introduced initiatives to foster Fintech innovation. SAMA's regulatory sandbox allows startups to test their solutions in a supervised environment. The CMA has issued digital financial services, crowdfunding, and cryptocurrency guidelines, promoting a diverse and secure financial sector.

United Arab Emirates: The United Arab Emirates (UAE) has emerged as a leading Fintech hub thanks to proactive regulatory measures by entities such as the Central Bank of the UAE (CBUAE) and the Dubai Financial Services Authority (DFSA). The CBUAE has developed regulations for digital payments, stored value facilities, and electronic KYC processes. The DFSA, operating in the Dubai International Financial Centre (DIFC), has established a Fintech regulatory framework with a sandbox for innovative solutions. Abu Dhabi Global Market (ADGM) also offers a robust Fintech regulatory regime, supporting a wide array of digital financial services.

2.2.3 The World Bank and International Organizations. The table below, derived from the World Bank's Global Fintech Enabling Regulations Database, provides a detailed overview of the regulatory frameworks across various fintech-related categories in Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates (UAE). It highlights the specific legislation and regulatory measures implemented in each country, covering critical areas such as anti-money laundering, equity crowdfunding, digital identity, central bank digital currencies (CBDCs), peer-to-peer (P2P) lending, electronic money, customer due diligence (CDD), cybersecurity, data protection, cryptocurrency, electronic payments, innovation

facilitators, digital banking, and open banking. This comprehensive comparison underscores the diversity in regulatory approaches within the GCC, offering insights into each nation's strategy to balance technological advancement with robust security and regulatory oversight.

2.3 International Telecommunications Union

Similarly, the International Telecommunications Union (ITU) ITU offers a variety of platforms to study cybersecurity challenges for digital finance, as well as innovations and regulatory challenges brought about by technologies like blockchain and artificial intelligence. [12](ITU, 2024). The ITU [13] (2004) also provides 'Key Recommendations for Digital Financial Services security', including security recommendations to protect against SIM risks, mobile security best practices, as well as a template for MOU between telecommunications regulator and Central Bank on Digital Financial Services (DFS) security. The last recommendation highlights the value of cooperation among different regulators, especially telecommunication and financial regulators. The usefulness of cooperation is highlighted later in the framework. The ITU also has a dedicated system, the Knowledge Sharing Platform, to foster collaboration among regulators and other stakeholders of security guidelines and best practices for DFS. It should also be noted that the ITU thought the Security Lab for Digital Financial Services assists regulators with DFS best practices. Therefore, there is regulation, best practices and mechanisms for cooperation at the international level. Nevertheless, the idiosyncrasies of GCC economies and, most importantly, Islamic finance will likely require more efforts to overcome Fintech barriers.

2.4 ISACA perspective

ISACA, the Information Systems Audit and Control Association is a globally recognised professional association dedicated to IT governance, risk management, and cybersecurity. Through its extensive network of industry experts and comprehensive research initiatives, ISACA provides valuable insights and guidelines for the evolving context of financial technology (Fintech). Among its numerous publications, three reports stand out as particularly relevant for understanding the cybersecurity challenges and risk management in fintech enterprises, and they offer a thorough examination of

Category	Bahrain	Kuwait	Oman	Qatar	Saudi Arabia	UAE
Anti Money Laundering	Decree-Law No. 4 of 2001	Anti-Money Laundering and Combating The Financing of Terrorism Law No. 106 of 2013	No. 30/2016	State of Qatar Anti-Money Laundering Act	Saudi Arabia's Anti-Money Laundering Law of 2003	Federal Decree-Law No. 20 of 2018 on Anti-Money Laundering and Combating the Financing of Terrorism and Financing of Illegal Organisations
Equity Crowdfunding	Regulated under the framework created for this activity	No legislation identified	No legislation identified	Regulated under the pre-existing regulatory framework	Regulated under the framework created for this activity	Regulated under the framework created for this activity
Digital ID	Law No. 46 Identity Card Act 2006	No legislation identified	Royal Decree 66/99 Promulgating the Law on Civil Status, Art. 42	Law No. 37 of 2005	Civil Status System, as amended by Council of Ministers Resolution No. 151 dated 13/5/1434H, Art. 67	Federal Law No. 9 of 2006, Art. 4
CBDC	No legislation identified	No legislation identified	No legislation identified	No legislation identified	No legislation identified	Development
P2P	Regulated under the framework created for this activity	No legislation identified	No legislation identified	No legislation identified	No legislation identified	Regulated under the framework created for this activity
Electronic Money	Legislative Decree No. 28 of 2002	No legislation identified	No legislation identified	No legislation identified	Payment Services Provider Regulations	Decretal Federal Law No. 14 of 2018
CDD	Decree-Law No. 4 of 2001	Anti-Money Laundering and Combating The Financing of Terrorism Law No. 106 of 2013	No. 30/2016	State of Qatar Anti-Money Laundering Act	Saudi Arabia's Anti-Money Laundering Law of 2003	Federal Decree-Law No. 20 of 2018 on Anti-Money Laundering and Combating the Financing of Terrorism and Financing of Illegal Organisations
Cyber Security	Decree-Law No. 28 of 2002	Law No. 63 of 2015 on Combating Cyber Crimes	Anti Cybercrime Act	Cybercrime Law	Anti Cyber Crime Law Royal Decree No. M/17	Federal Law by Decree No. 5 of 2012 regarding Cyber Crimes
Data Protection	Law No. 30 of 2018 regarding issuing the Law on the Protection of Personal Data	Law No. 20 of 2014 (the E-Commerce Law)	Electronic Transaction Law 2008	Law No. 13 of 2016 Concerning Personal Data Protection	Bill – Freedom of Information and Protection of Private Data	Dubai Law No. 26 of 2015 on the Regulation of Data Dissemination and Exchange in the Emirate of Dubai (“Dubai Data Law”)
Crypto Currency	No legislation identified	No legislation identified	No legislation identified	No legislation identified	No legislation identified	No legislation identified

Electronic Payment/Transactions	Decree-Law No. 28 of 2002	Law No. 20 of 2014 Concerning Electronic Transactions	Electronic Transactions Law	Decree-Law No. 16 of 2010 Electronic Commerce and Electronic Transactions Law	Electronic Transactions Law	Electronic Transactions and Commerce Law No. 2 of 2002, Federal Law 1/2006 Concerning Electronic Transactions and Commerce Law
Innovation Facilitators	Sandbox	Sandbox	No legislation identified	No legislation identified	Sandbox	Sandbox
Digital Banking	Regular Banking	Regular Banking	Regular Banking	Regular Banking	Regular Banking	Regular Banking
Open Banking	QB Open Banking Module	No legislation identified	No legislation identified	No legislation identified	No legislation identified	No legislation identified

Source: <https://www.worldbank.org/en/topic/fintech/brief/global-fintech-enabling-regulations-database>

the risks introduced by disruptive technologies, the complexities of maintaining robust cybersecurity frameworks, and innovative approaches to fintech risk assessment.

[15] Mani (2019) discusses the intersection of cybersecurity and Fintech, highlighting the new risk factors introduced by disruptive technologies. Fintech innovations, such as mobile payments and open banking, provide significant benefits but pose cybersecurity risks due to integration challenges, data privacy concerns, and the increasing sophistication of cyberattacks. The report emphasises the importance of robust third-party risk management, malware protection, and data leakage prevention. It also underscores the need for continuous cybersecurity awareness and training to mitigate these risks effectively.

[16] Stepanyan's (2022) report outlines the complexities of cybersecurity in Fintech enterprises, stressing the importance of timely breach detection and robust IT governance. It highlights the increasing threat landscape and the need for enterprises to adopt comprehensive cybersecurity frameworks like NIST CSF, COBIT, and ISO 27001. The report also discusses the critical role of leadership in fostering a cyber-resilient culture and the significance of continuous training and risk assessment in maintaining effective cybersecurity measures.

Finally, [14] Alvarez-Dionisi, L. E. (2020) presents a model for assessing risks in Fintech enterprises, integrating risk management frameworks and innovative solutions.

2.5 Other Organisations

The literature on the relationship between Fintech and cybersecurity focuses on threats. [17] Javaheri et al. (2024) provide a systematic literature review and in-depth analysis of cybersecurity threats in FinTech, offering insights for various stakeholders ranging from financial institutions to government agencies. They analyse 74 papers, including three other literature reviews and a wide range of cyber threats. In addition, the number of cyber incidents is growing, and it is imperative to understand the several threat categories posing severe cyber challenges to the FinTech industry worldwide [18] (Kaur et al., 2021).

Nevertheless, the issue of regulation is rarely mentioned. While there was a substantial increase in financial regulations introduced by the government and financial institutions in response to the financial crisis, it is suggested that regulatory bodies should revise and impose compliance on financial firms to adapt financial technologies in an ever-changing multi-jurisdictional regulatory context [17] (Javaheri et al., 2024). [19] Mehrban et al. (2020) found that regulatory requirements are an emerging FinTech issue and argue that regulating and controlling compliances is a fundamental aspect of FinTech. Another earlier literature review on the global regulatory environment and cybersecurity risks related to Fintech's complementary regulatory policies is suggested [20] (Ng and Kwok, 2017). Therefore, the literature has a lacuna that further justifies our framework and the focus on regulation of FinTech and cybersecurity [21] (Petratos and Faccia, 2019).

Despite the importance of cybersecurity regulation and related policies in Fintech, there seems to be sufficient research. [18] Kaur et al. (2021) propose a FinTech Information Security Governance framework to secure critical data from cyber-attacks. While this framework recognises the role of regulation, standards and other policies and mentions regulations, notably NIST 800-12, it provides a general guide. While some studies focused on certain cybersecurity threats, others have addressed the regulatory frameworks for FinTech cybersecurity; researchers have noted that these frameworks may not address FinTech cybersecurity concerns [3] (AlBenjasim et al., 2023). Our paper's framework has a more extensive and detailed scope, presenting regulations and policies and specific countries' weaknesses in practice. In addition, our framework does not apply in one country (i.e. [3] AlBenjasim et al., 2023 analysing Bahrain, etc.) but rather applies to a geographical area with idiosyncratic characteristics and highlights more complex issues as a country and international cooperation.

Category	Regulatory Needs	GCC Countries' Weaknesses	Factors for Industry Standards
Cybersecurity Regulations	ISO/IEC 27001, PCI-DSS	Inconsistent regulations	Multi-factor authentication, encryption
Regulatory Sandboxes	Support for fintech innovation	Cybersecurity skill gaps	Secure software development lifecycles
Cryptocurrency Regulations	Dual financial and Sharia regulations	High operational and compliance costs	Continuous monitoring and risk management
Collaboration and Innovation	Partnerships with regulators and experts	Variations in compliance requirements	Collaboration facilitated by ISACA
Emerging Technologies And other policies	Leveraging blockchain and AI	Lack of comprehensive fintech education	Blockchain for transparency and security

3 Framework and ANALYSIS

3.1 Framework matrix for enhancing security and performance in GCC Fintech

The framework matrix we propose is aimed at enhancing security in GCC fintech. It combines GCC countries' weaknesses, regulatory needs, factors for industry standards and best practices, and integration of technologies and other policies, offering a holistic approach to addressing the critical aspects of the fintech ecosystem.

Regulatory needs are centred on implementing robust cybersecurity regulations, such as ISO/IEC 27001 and PCI-DSS, exemplified by Bahrain's Central Bank requirements. Establishing regulatory sandboxes, like Saudi Arabia's SAMA sandbox, supports fintech innovation while managing risks. Dual regulation for cryptocurrencies ensures compliance with financial and Sharia principles, addressing the ambiguities in their adoption within Islamic finance.

GCC countries face specific weaknesses, including inconsistent regulatory frameworks across the region, which create compliance challenges. For instance, variations in the regulatory approaches of Bahrain and Saudi Arabia exemplify this inconsistency. Significant cybersecurity skill gaps are highlighted by the need for comprehensive cybersecurity education in GCC universities, as Abdeldayem and Al Dulaimi (2020) pointed out. Additionally, the high operational and compliance costs associated with maintaining cybersecurity measures pose a challenge, particularly for smaller startups in Saudi Arabia, as noted by Alghamdi and Elmedany (2023).

Comprehensive cybersecurity measures must be adopted to establish industry standards, including multi-factor authentication, encryption, and secure software development lifecycles. These best practices, as adopted by fintech firms in Bahrain, are crucial. Collaboration and innovation are also essential, with partnerships between fintech companies, regulators, and cybersecurity experts enhancing security measures. Continuous monitoring and risk management practices, such as regular risk assessments and data protection policies, are necessary to protect against evolving threats, as Bahrain's regulatory framework requires. Lastly, integrating emerging technologies like blockchain and artificial intelligence can significantly improve cybersecurity and operational efficiency, as evidenced by the potential of blockchain technology to enhance transparency and security in financial transactions [7] (Allen, 2021).

This interconnected framework emphasises the need for regulatory harmonisation, educational improvements, and the adoption of best practices and emerging technologies to secure and enhance the performance of the GCC fintech sector.

3.2 Framework Matrix for Enhancing Security and Performance in GCC Fintech: Emerging Technologies

Integrating emerging technologies in the GCC fintech sector enhances security and performance. This framework matrix combines regulatory needs, identified weaknesses within GCC countries, and essential factors for establishing industry standards, thus effectively addressing the claims of the title.

Regulatory needs focus on developing comprehensive frameworks for blockchain technology, ensuring ethical use of AI and machine learning, and implementing clear regulations for digital currencies and tokens. For instance, Bahrain's regulatory requirements for fintech companies to adopt extensive cybersecurity measures illustrate the importance of robust cybersecurity regulations. Additionally, Saudi Arabia's SAMA regulatory sandbox supports fintech innovation while managing potential risks, showcasing the benefits of regulatory sandboxes. The dual regulation of cryptocurrencies, addressing financial and Sharia compliance, highlights the need for clear guidelines.

GCC countries face several weaknesses, including inconsistent regulatory frameworks that challenge compliance. For example, the varying regulatory approaches between Bahrain and Saudi Arabia exemplify these inconsistencies. Technological infrastructure gaps, such as limited access to high-speed internet and advanced computing resources, hinder the widespread adoption of emerging technologies. There are also significant skill gaps in blockchain, AI, and other advanced technologies, necessitating specialised education and training programs.

To establish industry standards, it is essential to adopt comprehensive cybersecurity measures like multi-factor authentication and encryption, as demonstrated by best practices in Bahrain. Collaboration between fintech companies, regulators, and cybersecurity experts is crucial for enhancing security measures. Continuous monitoring and risk management practices, such as regular risk assessments and data protection policies required by Bahrain's regulatory framework, are necessary to protect against evolving threats. Leveraging blockchain and artificial intelligence can significantly

Category	Regulatory Needs	GCC Countries' Weaknesses	Factors for Industry Standards
Blockchain Technology	Regulatory frameworks for blockchain	Lack of standardised frameworks	Secure transaction records through blockchain
AI and Machine Learning	Guidelines for ethical AI use	Technological infrastructure gaps	AI for real-time fraud detection
Digital Currencies and Tokens	Regulations for digital currencies and tokens	Skill gaps in emerging technologies	Digital identity solutions for secure user authentication
Technological Infrastructure	Investment in technological infrastructure	Insufficient support for emerging technologies	Advanced data analytics for strategic decision-making
Education and Training	Specialised programs for blockchain and AI	Lack of skilled professionals	Cybersecurity innovations to protect against cyber threats
Collaborative Efforts	Policies promoting collaboration between regulators and industry	Inconsistent regulatory approaches	Collaboration to standardise blockchain and AI applications

improve cybersecurity and operational efficiency, as evidenced by blockchain's potential to enhance transparency and security in financial transactions.

The matrix integrates regulatory needs, weaknesses, and industry standards to form a cohesive framework. It emphasises the need for harmonised regulations, improved technological infrastructure, and the development of a skilled workforce to support the integration of emerging technologies in the fintech sector. Collaborative efforts and continuous innovation are key to securing and enhancing the performance of GCC fintech.

3.2.1 Emerging Technologies and Other Policies.

4 Conclusions

The research into enhancing security and performance in GCC fintech has revealed critical insights into the regulatory frameworks, industry standards, and emerging technologies shaping the sector. By leveraging the extensive resources and expertise of ISACA and ACM, this study has provided a comprehensive analysis that bridges the gap between academic research and industry practice. The integration of robust cybersecurity measures, innovative technologies, and collaborative efforts between stakeholders is essential for the sustainable growth of Fintech in the GCC region.

Key findings highlight the importance of implementing comprehensive cybersecurity frameworks, such as ISO/IEC 27001 and PCI-DSS, and the benefits of regulatory sandboxes that support fintech innovation while managing risks. The dual regulation of cryptocurrencies, ensuring compliance with financial and Sharia principles, addresses significant ambiguities in adopting digital currencies within Islamic finance.

GCC countries face several challenges, including inconsistent regulatory frameworks, technological infrastructure gaps, and skill shortages in emerging technologies. Addressing these weaknesses requires harmonised regulations, improved technological infrastructure, and the development of specialised education programs to foster a skilled workforce.

Adopting emerging technologies, such as blockchain and artificial intelligence, offers significant potential to enhance security and operational efficiency in Fintech. Collaborative efforts between

fintech companies, regulators, and cybersecurity experts are crucial for establishing industry standards and ensuring continuous monitoring and risk management practices.

4.1 Limitations

Despite the comprehensive approach, this research has certain limitations. Firstly, the rapidly evolving nature of Fintech and cybersecurity means that the findings may quickly become outdated as new technologies and regulations emerge. Additionally, the study's focus on the GCC region may limit the generalizability of the findings to other regions with different regulatory environments and market dynamics. The reliance on qualitative data from expert opinions and case studies may also introduce subjective biases, affecting the overall conclusions.

4.2 Future Research

Future research should continuously monitor fintech developments and regulatory changes to keep the findings relevant and up-to-date. Expanding the scope to include comparative studies with other regions could provide a broader perspective on global fintech trends and regulatory best practices. Further quantitative research, including large-scale surveys and empirical data analysis, can enhance the robustness of the findings and provide more objective insights into the impact of emerging technologies and regulatory measures on fintech security and performance.

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